

X40Y Wide Area Network HTTP Interface Protocol

V1.01

Catalogue

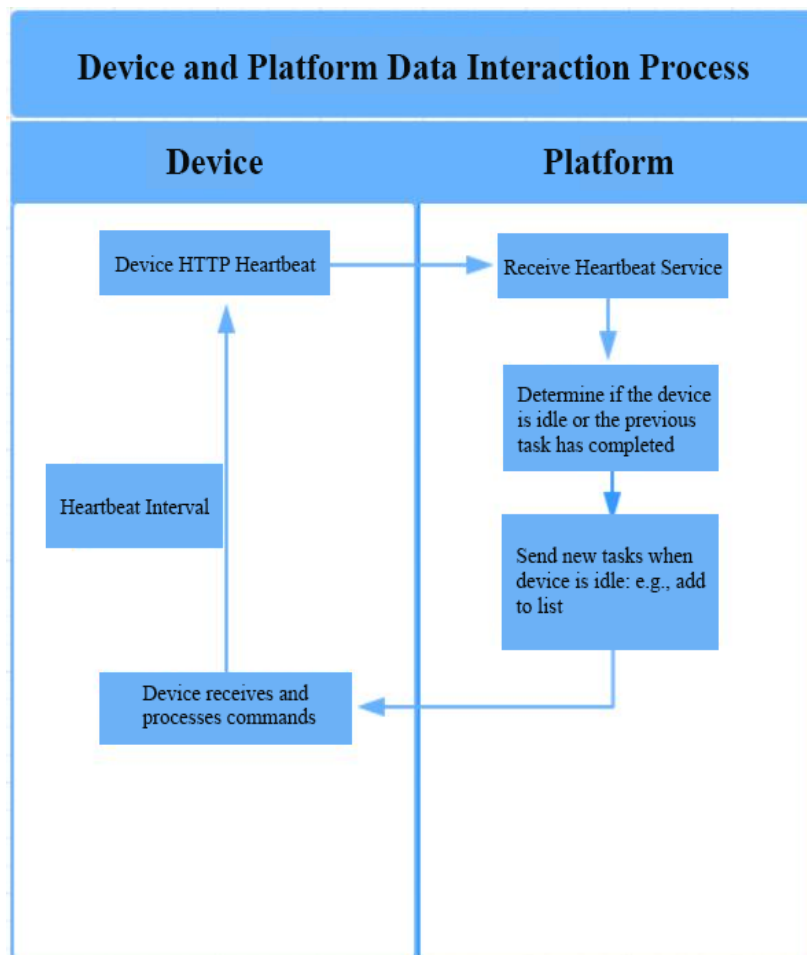
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1. Summary

This document mainly refers to the HTTP interface protocol V1.04.pdf of the Face Recognition All-in-One Machine for wide area network, which is used for the **wide area network** HTTP docking of network cameras, and is suitable for system development and maintenance personnel.

2. Principle of Interface

Since **network cameras** (hereinafter referred to as: **devices**) are all deployed in the Intranet, the platform cannot access the network cameras in the Intranet through the Extranet, so the WAN interface is based on **HTTP heartbeat** to achieve data interaction.
tree :



2.1 Request Method

The request uses the POST method, with the data for sending and uploading stored in the entity-body.

2.2 Data Types and Formats

2.2.1 All strings should be encoded in UTF-8.

2.2.2 The data exchanged between both parties is uniformly packaged in JSON format, with the basic format as follows:

```
{
  "operator": "<Operator>",
  "info": {<Info>}, .....
}
```

For more details, see the specific data format of each interface.

2.2.3 All image data uses base64 encoding or image URI.

2.3 Grammar

The requested URI format is: `http://<server ipaddr>/action/<Operator>`. Here, `<server ipaddr>` is the IP address of the network camera you are requesting, and `<Operator>` is the action content. For details, refer to the specific interface.

3. Device Heartbeat

The device retrieves the heartbeat address, port, and URL from the data subscription information. Throughout the subscription validity period, the device sends heartbeat messages to the subscription address. The subscription server must respond with a heartbeat to maintain a valid connection. If the heartbeat handshake fails, the device will disconnect and reconnect to the subscription address. The heartbeat interval is set to 30 seconds. Heartbeat reporting requires any HTTP subscription (authenticated or non-authenticated, including stranger capture subscriptions), and cannot be subscribed to independently. Modify the heartbeat interface based on the LAN HTTP interface documentation to enable data interaction between devices and the platform. **Note: Red font indicates new parameters added outside the LAN HTTP interface.**

Parameter information (Note: optional):

Key	Type	Values	Description
operator	char	HeartBeat	Heartbeat
info	json object		Concrete content
DeviceID	int		Device ID of the device to be operated
Time	char	YYYY-MM-DDThh:m m:ss	The time at which the device was placed e. g. , 2018-03-25T21:50:05
data	Json object		Command and execution data

StateCode	int		When StateCode is 0, the device is idle or the task is complete. If the device is idle, it returns the command operator = "none"; If the device completes its previous task, it returns the corresponding operator command, including the TaskId of the returned task and the result of the previous command operation. When StateCode is 1, it indicates the device is operational and returns the task ID and current completion status.
Taskid	Int		Running or last task ID
OperatorInfo	Json object		Instruction information
operator	char		Order set none AddPerson 3.10.2 Add Person to multiple lists AddPerson 3.10.1 Add Person to a single list
SuccessCount	int		Number of successful attempts
ProgressText	char		Task progress text such as 100/1000
errorInfo	Json object		Error message feedback
errorItem	char		Error item
errordesc	char		Cause of error

e.g., device heartbeat reporting

```
{
  "operator": "HeartBeat",
  "info": {
    "DeviceID": 1300001,
    "Time": "2018-03-12T09:09:20"
  },
  data ": { // request content"
    TaskId": 1, // task ID
    StateCode: 0, //Device status: 0-idle/completed, 1-in operation
    OperatorInfo": { //instruction information
      operator ":" addPersons", // instruction: AddPersons is the LAN HTTP interface
document 3.10.2 for adding multiple lists
      SuccessCount: 1000, //Number of successful entries
      ProgressText ":" 100/1000", //Task Progress
      "errorInfo":{
        errorItem: "4476c20c-23ce-4178-8672-b292d33a3cd6", //error item such as personnel ID

```

```

        "errordesc": "Face not va-
lid"}, {
        errorItem: "5376c20c-23ce-4178-8672-b292d33a3cd6", //error item such as
        personnel ID "errordesc": "Face not qualified"
    }]
}
}
}

```

e.g. the platform issues a batch add list command

```

{
    "Stamp": 1555598153, // timestamp
    "TaskId": 1, //Task ID
    "OperatorInfo": { // command information
        operator ":" AddPersons", // instruction: AddPersons is the LAN HTTP interface
        document 3.10.2 for adding multiple lists
        "DeviceID": 1300001,
        "Total": 3,
        "Personinfo_0": { "
            PersonType": 0,
            "Name": "test",
            "Gender": 0,
            "Nation": 0,
            "CardType": 0,
            "IdCard": "430923199011044411",
            "Birthday": "1990-11-04",
            "Telnum": "18888888888",
            "Native": "Guangdong",
            "Address": "Shenzhen Bao'an",
            Notes: "Notes"
            "MjCardFrom": 2,
            "WiegandType": 7,
            "CardMode": 1,
            "WGFacilityCode": 9 A,
            "MjCardNo": 0192B3,
            "RFIDCard": "9A0192B3",
            "Tempvalid": 0,
            "CustomizeID": 120,
            "PersonUUID": "4476c20c-23ce-4178-8672-b292d33a3cd6",
            "isCheckSimilarity": 0,
            "ValidBegin": "2019-02-12T09:09:20",
            "ValidEnd": "2019-03-12T09:09:20"
            , "picURI": "https://btngps.oss-cn-beijing.aliyuncs.
com/image/xxx.jpg"
        }
    }
}

```

```

    },
    "Personinfo_1": {
        .....
    }
    .....
}
}
}

```

4. WAN Support Instruction Set

The following is the interface instruction set supported by WAN, which will be different from the LAN version. For the specific inner data structure, see [Network Camera Http Interface Protocol Vx.xx](#), where x.xx is the version number of the specific protocol description.

3.1 Add and edit the list Add-Persons and EditPersonsNew

The AddPersons interface supports adding up to 500 personnel when the list image is added as a URI, and the EditPersonsNew interface supports adding up to 500 personnel when the list image is added as a URI. The AddPersons and EditPersonsNew interfaces cannot be executed simultaneously.

3.2 Delete list

DeletePerson DeleteAllPerson

You cannot execute DeletePerson, AddPersons, and EditPersonsNew simultaneously.

3.3 Data Subscription

Subscribe, SnapPush, VerifyPush, GetSubscribe

3.12.9 Query subscribed information

3.4 Unsubscribe

3.5 Heartbeat Subscription HeartBeat

3.6 Search by List

SearchPerson SearchPersonList

5. Example of Overall Data Interaction Process

1. The device sends heartbeat to the platform server

```
POST /crane/heartbeat/1413724 HTTP/1.1
Accept: application/json
Content-Type: application/json; charset=UTF-8
Content-Length: 340
Accept-Encoding: gzip, deflate
User-Agent: FaceGate-HttpClient/0.0.1
Host:192.168.0.22
Connection: Keep-Alive
```

```
{
  "operator": "HeartBeat",
  "info": {
    "DeviceID": 1413724, "Time": "
2020-12-14T20:46:56"}, "data":
{
  "TaskId": 0,
  "StateCode": 0,
  "OperatorInfo":{
    "operator": "none",
    "SuccessCount": 0,
    "ProgressText": "",
    "ErrCount": 0,
    "errorInfo": [{
      "errcode":"0",
      "errorItem":"111",
      "errordesc": "Idle",
    }],
    "SuccessInfo":[]
  }
}
```


}

2. The platform responds with a heartbeat and adds instructions (e.g., batch add personnel, add personnel via URI. This interface can add up to 500 personnel at once).

HTTP/1.1 200 OK
Server: nginx/1.18.0
Content-Type: application/json
Content-Length: 362
Connection: keep-alive
X-Powered-By: PHP/7.4.9
Date: Mon, 14 Dec 2020 12:40:34 GMT

```
{"TimeStamp":1607949634,"TaskId":1,"OperatorInfo":{"Total":1,"DeviceID":"1413724","operator":
":"AddPersons","Personinfo_0":{"PersonType":0,
      "Name":"test0",
      "CustomizeID": 120,
      "PersonUUID": "4476c20c-23ce-4178-8672-b292d33a3c20",

"picURI":"https://hqtestvt.oss-cn-shenzhen.aliyuncs.com/faceTest/f3c0bb1e-c703-48b1-a79f-a9c
2efe2axxx.jpg"}}
```

3. The device returns the result of the previous command data operation on the next heartbeat response platform.

If the device is idle, the platform can proceed to step 2. If the device has not completed the previous command action, the platform's second step will set "operator" to "none".

POST /crane/heartbeat/1413724 HTTP/1.1
Accept: application/json
Content-Type: application/json; charset=UTF-8
Content-Length: 365
Accept-Encoding: gzip, deflate

User-Agent: FaceGate-HttpClient/0.0.1

Host:192.168.0.22

Connection: Keep-Alive

```
{
  "operator": "HeartBeat",
  "info": {
    "DeviceID": 1413724, "Time": "
2020-12-14T20:47:25"
  },
  "data":{
    "TaskId": 1,
    "StateCode": 0,
    "OperatorInfo":{
      "operator": "AddPersons", "SuccessCount
": 1, "ProgressText": "1/1", "ErrCount": 0,
      "errorInfo": [], "SuccessInfo":{"
Personinfo":0, "CustomizeID":121, "
PersonUUID":" 130922xxxx_2020-12-12"
    }}
  }
}
```

4. The previous heartbeat command has been completed. Next, send the command data (add another person).

HTTP/1.1 200 OK

Server: nginx/1.18.0

Content-Type: application/json

Content-Length: 364

Connection: keep-alive

X-Powered-By: PHP/7.4.9

Date: Mon, 14 Dec 2020 12:40:34 GMT

```
{"TimeStamp":1607949634,"TaskId":1,"OperatorInfo":{"Total":1,"DeviceID":"1413724","operator
":"AddPersons","Personinfo_0":{"PersonType":0,
  "Name":"test123",
```

"CustomizeID": 121,

"PersonUUID": "4476c20c-23ce-4178-8672-b292d33a3c21",

"picURI": "https://hqtestvt.oss-cn-shenzhen.aliyuncs.com/faceTest/f3c0bb1e-c703-48b1-a79f-a9c2efe2xxxx.jpg"}}